

Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 – Artificial Intelligence and Machine Learning

Assignment Specification

Assignment Title

Design, Implement, and Evaluate AI/ML Solutions for a Real-World Problem Using a Given Dataset

Objective

To design and implement an end-to-end AI/ML solution for a real-world problem using suitable supervised or unsupervised learning algorithms, applied to a dataset assigned to you, and to critically evaluate the performance, effectiveness, and ethical implications of the chosen approach.

Learning Outcomes Addressed

LO3: Apply AI/ML techniques to solve real-world problems.

LO4: Critically evaluate the effectiveness and impact of AI/ML algorithms.

Assignment Components

Step 1: Assigned Dataset Understanding and Problem Framing (10 Marks)

- Each student will be provided with a specific dataset by the module coordinators. Dataset selection is not required this year. Your task is to understand, describe, and justify the suitability of your assigned dataset.
 - Identify a real-world problem domain such as healthcare, finance, agriculture, education, or another relevant area that aligns with the assigned dataset.
 - Describe and justify the dataset's suitability by considering its features, labels or target variable, size, structure, and overall data quality.
 - Clearly define the AI/ML problem to be addressed, such as classification, regression, clustering, or another appropriate task.
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- Briefly state which AI tools, if any, you plan to use during the project and explain their intended purpose. Refer to the AI Tool Usage Declaration section below.

Step 2: Data Preprocessing and Analysis (25 Marks)

- Clean and preprocess the assigned dataset. This may include handling missing values, duplicate records, outliers, categorical variables, encoding, feature scaling or normalization, and other dataset-specific data quality issues.
- Perform exploratory data analysis using suitable statistical methods and visualizations.
- Examine relevant feature distributions, relationships, correlations, class distributions, and other meaningful patterns.
- Provide clear interpretations of the insights and patterns observed in the data.
- Explain how findings from the exploratory data analysis influenced subsequent preprocessing and modelling decisions.

Step 3: Model Design and Implementation (25 Marks)

- Select and justify at least two AI/ML algorithms suitable for the assigned dataset and problem.
- Examples may include Logistic Regression, Support Vector Machines, Decision Trees, Random Forests, K-Nearest Neighbours, K-Means Clustering, Neural Networks, or other relevant machine learning or deep learning approaches.
- Implement the selected models using Python and appropriate libraries or frameworks such as Scikit-learn, PyTorch, TensorFlow, or equivalent tools.
- Clearly document the modelling workflow and important implementation choices.
- Include appropriate model pipeline steps.
- Perform hyperparameter tuning using a suitable approach such as Grid Search, Random Search, or another justified method.
- Ensure that data preprocessing and model training are conducted in a manner that avoids data leakage.

Step 4: Model Evaluation and Comparison (30 Marks)

- Evaluate each model using performance metrics appropriate to the selected problem.
 - Depending on the task, suitable metrics may include Accuracy, Precision, Recall, F1 Score, Confusion Matrix, ROC-AUC, Mean Absolute Error, Mean Squared Error, R^2 Score, Silhouette Score, or other relevant evaluation measures.
 - Perform k-fold cross-validation where appropriate to assess the reliability and generalizability of the models.
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- Present and interpret the evaluation results using suitable tables and visualizations.
- Compare the selected models critically rather than relying only on a single performance metric.
- Identify the best-performing model and provide a clear justification for your selection.
- Discuss potential limitations, overfitting or underfitting issues, and opportunities for further improvement.

Step 5: Individual Report (10 Marks)

- Prepare a professional individual report with a maximum length of 15 pages.
- The report should contain the following sections:
 1. Introduction and Problem Statement
 2. Dataset Description, including source, structure, features, target variable or labels, and dataset characteristics
 3. Data Preprocessing and Exploratory Data Analysis
 4. Model Design and Implementation
 5. Model Evaluation and Comparison
 6. Ethical Considerations and Bias Mitigation
 7. AI Tool Usage Declaration and Transparency
 8. Individual Reflections and Lessons Learned
 9. References
- Submit the final report in PDF format before the viva session or the stated submission deadline.
- The quality and relevance of visualizations, technical accuracy, clarity of explanations, logical organization, professional presentation, and evidence of independent understanding and decision-making will also be evaluated.

Ethical Considerations and Bias Mitigation

- As part of the project, you should critically consider the ethical implications of your AI/ML solution.
- Relevant issues may include bias or imbalance within the dataset, fairness of model predictions, privacy and confidentiality, potential misuse of predictions, explainability and transparency, consequences of false positives and false negatives, limitations of the dataset, and risks associated with deploying the model in the selected real-world domain.
- Where relevant, explain any steps taken to identify, reduce, or manage these risks.

AI Tool Usage Declaration and Transparency

- Students must be transparent about how generative AI tools such as ChatGPT, Claude, GitHub Copilot, Gemini, or similar tools were used during the project.
- In the dedicated report section, include a short table or written declaration covering each AI tool used. For each tool, include the tool name and version, if known.
- Explain the specific aspects supported by the tool. Examples may include code debugging, exploratory data analysis or visualization ideas, hyperparameter tuning suggestions, model selection ideas, explaining algorithms, brainstorming problem framing, report writing or editing, improving clarity or grammar, and generating documentation.
- State the extent of use for each aspect, for example minimal, moderate, or extensive. Explain how you verified, tested, modified, or validated AI-assisted outputs before including them in your project.
- You remain responsible for understanding, verifying, and taking ownership of all code, analysis, interpretations, and written content submitted as part of the project.
- Students who did not use any AI tools should simply state this clearly. Honest disclosure is what is being assessed, not the amount of AI use itself.

Individual Project Expectations

- As this is an individual project, all data analysis, modelling, evaluation, interpretation, and report preparation must reflect your own understanding and individual contribution.
- You should be able to explain and justify all preprocessing steps, algorithm choices, hyperparameters, evaluation metrics, and conclusions during the viva.
- Any external resources, code examples, datasets, academic literature, or AI-assisted material used should be appropriately acknowledged where required.
- The submitted work should demonstrate independent problem-solving, critical thinking, and technical understanding.